

Neurosciences and behavior module(NSBE 2131)

(Dec 4 ,2023 -Jan 19, 2024), ECTS =10 (~116 hrs)

Coordinating Department: **Pharmacology**

Participating Departments: **Anatomy, Biochemistry, Physiology, DMIP, Pathology and Pharmacology**

Coordinator: **Mr Samuel Worku**

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Office: New building, fifth floor room number 510

Anatomy
lecture

Physiology
lecture

Biochemistry
lecture

Microbiology
lecture

Pathology
lecture

Pharmacology
lecture

Week One

TIME	MONDAY Dec 4	TUESDAY Dec 5	WEDNESDAY Dec 6	THURSDAY Dec 7	FRIDAY Dec 8
9:00:-12:00AM		DR 1: Group 1 DR 2: Group 2 Mr. Fikre Mrs. Shewit	DR 1: Group 3 DR 2: Group 4 Mr. Assaye Dr. Solomon	DR 1: Group 5 DR 2: Group 6 Mr. Solomon Mr. Yossef	DR 1: Group 1 DR 2: Group 2 Mr. Abay Mr. Fikre
			Hist. Lab 1: Gr. 1 Hist Lab 2: Gr. 2 Dr. Soressa Abebe	Hist. Lab 1: Gr. 3 Hist Lab 2: Gr. 4 Mrs. Shewit	Hist. Lab 1: Gr. 5 Hist Lab 2: Gr. 6 Mr. Assaye
12:00-1:30 PM	<i>Lunch</i>				
1:30-3:00 PM	Introduction and Organization of the Nervous System (CNS) and parts of the brain Cranial meninges and dural venous sinuses , and cisterns Prof. Girma Seyoum	<i>Neurohistology</i> <i>Microscopic structures of the CNS</i> Dr. Soressa Abebe	<i>Neuroanatomy</i> Midbrain: (tectum colliculi, tegmentum; cerebral peduncles; structure, nuclei & tracts The base of the brain (interpeduncular fossa and superficial attachments of cranial nerves, and circle of Willis. Transverse section of the Brain stem and cerebellum Prof. Girma Seyoum	<i>Neuroanatomy</i> Brain by Sections: Sagittal (corpus callosum, fornix septum pellucidum fornix, thalamus, hypothalamus and 3rd ventricle and etc.) Horizontal (Gray matter of cerebrum, , basal ganglia internal capsule and corpus striatum and etc.) Coronal (Lateral , insula, ventricle, white matter of cerebrum) Prof. Girma Seyoum	<i>Neuroanatomy</i> Cranial nerves I: Purely sensory (I, II & VIII). Purely motor (III, IV, VI & XII), their pathways; Nuclei: Location & Function; Prof. Girma Seyoum
3:00-3:15 P.M.	<i>Break</i>				
3:15-10:30 P.M.	Hindbrain: Medulla oblongata and Vertebral Pons and Basilar artery. Prof. Girma Seyoum	<i>Neuroanatomy</i> Hindbrain: Cerebellum and 4th ventricle Prof. Girma Seyoum	<i>Neuroanatomy</i> Cerebrum I (cerebral hemisphere, cerebral cortex, lobes, gyri, sulci / fissures and The cerebrum II (cortical areas, and functional localization) Prof. Girma Seyoum	<i>Neuroanatomy</i> Spinal cord: spinal meninges and supports, spinal nerves and vertebral, Prof. Girma Seyoum	<i>Neuroanatomy</i> Cranial nerves II: Mixed cranial nerves (V, VII, IX & X), their pathways; Nuclei: Location & Function; Prof. Girma Seyoum

Week Two

TIME	MONDAY Dec 11	TUESDAY Dec 12	WEDNESDAY Dec 13	THURSDAY Dec 14	FRIDAY Dec 15
9:00:-12:00AM	DR 1: Group 3 DR 2: Group 4 Mrs. Shewit Mr. Yossef	DR 1: Group 5 DR 2: Group 6 Mr. Fikre Mr. Assaye	DR 1: Group 1 DR 2: Group Dr. Solomon Mr. Abay	DR 1: Group 3 DR 2: Group 4 Mr. Solomon Mr. Fikre	DR 1: Group 5 DR 2: Group 6 Mr. Abay Mr. Yossef
	Hist. Lab 1: Gr. 1 Hist Lab 2: Gr. 2 Mr. Solomon	Hist. Lab 1: Gr. 3 Hist Lab 2: Gr. 4 Mrs. Shewit	Hist. Lab 1: Gr. 5 Hist Lab 2: Gr. 6 Mrs. Shewit	Hist. Lab 1: Gr. 1 Hist Lab 2: Gr. 2 Mr. Assaye	Hist. Lab 1: Gr. 3 Hist Lab 2: Gr. 4 Mr. Assaye
12:00-1:30 PM	<i>Lunch</i>				
1:30- 3:00 P.M.	<i>Neuroanatomy</i> TRACTOLOGY I; Ascending tracts of the nervous system: Sensory pathways (Spinothalamic; Dorsal Column/Medial Lemniscal & Trigeminal Nerve Pathways/Tracts; Spino-Cerebellar Tracts) Prof. Girma Seyoum	Neurohistology Microscopic structures the Eye and Ear Dr. Soressa Abebe	Neurohistology Sensory system: Photoreceptor system Sensory system: Dr. Soressa Abebe	Neuroembryology: Development of the nervous system: CNS & Malformations Mr. Reta Wakoya	<i>Neuroembryology</i> Development of the eye and its congenital anomalies Mr. Reta Wakoya
3:00 – 3:15P.M.	<i>Break</i>				

3:15-4:30 P.M.	TRACTOLOGY II Descending tracts / Motor pathways of the nervous system (Corticospinal; Corticobulbar; Extrapyramidal Motor pathways), UMN & LMN Prof. Girma Seyoum	Neurohistology Audioreceptor system Other sensory organs Dr. Soressa Abebe	Neuroembryology: Development of the nervous system: CNS & Malformations Mr. Reta Wakoya	Neuroembryology: Development of ear and malformations Mr. Reta Wakoya	Clinical correlation: neurological examination Mr. Reta Wakoya
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Week Three

TIME	MONDAY Dec 18	TUESDAY Dec 19	WEDNESDAY Dec 20	THURSDAY Dec 21	FRIDAY Dec 22
9:00-12:00AM	DR 1: Group 5 DR 2: Group 6 Dr. Soressa Abebe Dr. Solomon Tibebe	DR 1: Group 1 DR 2: Group 2 Dr. Solomon Tibebe Mr. Assaye	DR 1: Group 3 DR 2: Group 4 Dr. Soressa Abebe Mrs. Shewit	DR 1: Group 1 DR 2: Group 2 Mr. Abay Mr. Fikre	DR 1: Group 3 DR 2: Group 4 Dr. Soressa Abebe Mrs. Shewit
	Hist. Lab 1: Gr. 3 Hist Lab 2: Gr. 4 Mrs. Shewit	Hist. Lab 1: Gr. 5 Hist Lab 2: Gr. 6 Dr. Soressa Abebe	Hist. Lab 1: Gr. 1 Hist Lab 2: Gr. 2 Mr. Assaye	Hist. Lab 1: Gr. 5 Hist Lab 2: Gr.6 Mr. Assaye	Hist. Lab 1:Gr. 1 Hist Lab 2: Gr. 2 Dr. Solomon Tibebe
12:00-1:30 PM					
1:30- 3:00 P.M.	Neurophysiology Functional organization of the cerebral lobes, brain Dr Mekoya	Thermoception and nociception Motor system ... Dr Mekoya	Physiology of hypothalamus Physiology of synapse Dr Abebaye	Higher brain functions: Dr Abebaye	Diffuse modulatory system.Somatosensory system: Receptors, afferent nerves, Dr Dres
3:00 – 3:15P.M.	<i>Break</i>				
3:15-4:30 P.M.	stem and spinal cord, Milieu of the CNS cerebral blood flow, and	: Motor system ... Dr Mekoya	Blood brain barrier and cerebrospinal fluid Dr Abebaye	Higher brain functions..... Dr Abebaye	Visual system... Dr Dres

	metabolism				
	Dr Mekoya				

Week Four					
TIME	MONDAY Dec 25	TUESDAY Dec 26	WEDNESDAY Dec 27	THURSDAY Dec 28	FRIDAY Dec 29
9:00-12: 00AM	DR 1: Group 5 DR 2: Group 6 Mr. Assaye Mrs. Shewit Hist. Lab 1: Gr. 3 Hist Lab 2: Gr. 4 Mr. Yossef	DR 1: Group 1 DR 2: Group 2 Dr. Soressa Abebe Mrs. Shewit Hist. Lab 1: Gr. 5 Hist Lab 2: Gr. 6 Mr. Assaye	DR 1: Group 3 DR 2: Group 4 Mr. Abay Dr. Solomon Tibebu	DR 1: Group 5 DR 2: Group 6 Mr. Abay Mr. Fikre Biochemistry Seminars Gr 1 and 2	
12:00-1:30 PM	<i>Lunch</i>				
1:30-3:00 PM	Audition: Olfaction, Olfactory receptors Dr Dres	Olfactory signal transduction; Gustation: Gustatory Dr Dres	<i>Biochemistry</i> The Biochemistry of neurotransmitter secretions-amino acids and their derivatives, Mr Muluken	The Biochemistry of neurotransmitter con'd Mr Muluken	Neroanatomy&Neur ophysiology Quiz1
3:00-3:15PM	<i>Break</i>				
3:15-4:30 PM	and their cellular organization,	receptors and their cellular organization,	The Biochemistry of catecholamines,	Biochemistrof progressive	

	Dr Dres	gustatory signal transduction Dr Dres	peptides, and gaseous neurotransmitters. Mr Muluken	neurodegenerative diseases Mr Muluken	
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Week Five					
TIME	MONDAY Jan 1	TUESDAY Jan 2	WEDNESDAY Jan 3	THURSDAY Jan 4	FRIDAY Jan 5
9:00:-12: 00 AM	DR 1: Group 1 DR 2: Group 2 Dr. Solomon Tibebe Mrs. Shewit Biochemistry Seminars Gr 3and 4	DR 1: Group 3 DR 2: Group 4 Mr. Assaye Mrs. Shewit Biochemistry Seminars Gr,5 and 6 Microbiology Practical Gr 1 and 2	DR 1: Group 5 DR 2: Group 6 Mr. Abay Mr. Assaye Microbiology Practical Gr 3 and 4 (practical) Group 1 and 2	DR 1: Group 1 DR 2: Group 2 Mr. Abay Mr. Yossef Pathology (practical) Group 3 and 4 Microbiology Practical Gr 5and 6	DR 1: Group 3 DR 2: Group 4 Mr. Solomon Mrs. Shewit Neuropharmacology Seminar Gr 1and 2
12:00-1:30 PM	<i>Lunch</i>				
1:30-3:00PM	<i>Microbiology</i> CNS infections Dr. Brhanu	Viral meningitis Fungalmeningitis Tetanus (<i>C. tetani</i>) Dr Brhanu	Encephalitis Polio (polio virus) Slow virus infections (conventional and non-conventional causes Prof. Monga	<i>Pathology</i> Congenital diseases of CNS & CNS Trauma Stroke & Dr Amanuel.	CNS tumorsCNS Infections: meningitis, Dr Amanuel.
		<i>Break</i>			
3:15 – 4:30PM	Bacterial meningitis Dr. Brhanu	Toxoplasmosis of CNS	Eye infections (Outer and inner eye)		

		Mr. Habtamu	Ear infections (Otitis interna, media and externa) Botulism (<i>C. botulinum</i>) Dr. Daniel	neurodegenerative disorders Dr Amanuel.	encephalitis Pathology of the eye Dr Amanuel.
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Week six					
TIME	MONDAY Jan 8	TUESDAY Jan 9	WEDNESDAY Jan 10	THURSDAY Jan 11	FRIDAY Jan 12
9:00-12:00 AM		DR 1: Group 5 DR 2: Group 6 Mr. Fikre Mrs. Shewit Neuropharmacology practicals Gr 1 and 2	neuropharmacology Seminar Gr 5 and 6	Neuropharmacology practicals Gr 3 and 4 Pathology (practical) Group 5 and 6	Neuropharmacology practical Gr 5 and 6 Neuropharmacology Seminar Gr 3 and 4
12:00-1:30 PM					
1:30- 3:00 P.M.		Pharmacology Classification of drugs acting on CNS General anesthetics Prof Eyasu	Alcohols, Antiepileptics Prof Eyasu	Antipsychotics, Mr Akeberegn	Analgesics :Opioids and non-opioids CNS stimulants, Mr Akeberegn

3:00 – 3:15 P.M.	ho	<i>Break</i>			
3:15-4:30 P.M.		General anesthetics... Sedative/hypnotics <i>Prof Eyasu</i>	Antiparkinsons drugs <i>Prof Eyasu</i>	Antidepressants Antimanic drugs <i>Mr Akeberg</i>	Social Pharmacology (Drug dependence and Addiction) <i>Mr Akeberg</i>

Week Seven/ Exam Week					
TIME	MONDAY Jan 15	TUESDAY Jan 16	WEDNESDAY Jan 17	THURSDAY Jan 18	FRIDAY Jan 19
9:00-12:00 AM	Biochem, Micro &Patho Quiz 2				
12:00-1:30 PM					
1:30-4:30 PM					

					Final Exam